University of KwaZulu–Natal

School of Mathematics, Statistics and Computer Science

COMP301 (2026)

Software Design

CONFIDENT AGENT

Designing for Trust under Uncertainty

"A Confident Agent is not one that is always correct.
It is one that knows when it should not be trusted."

This project investigates how thoughtful software design can improve the trustworthiness of AI systems through benchmark design, experimental methodology, validation architectures, and evidence-based evaluation. Students are challenged not merely to build an intelligent system, but to justify why it deserves to be trusted.

Group Project Assignment

Semester Two
2026

Version 1.0

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1 Research Principles

> *"The future of AI is not systems that never make mistakes, but systems that know when they might."*

Large Language Models have transformed modern software development, scientific research, and knowledge work. Despite their remarkable capabilities, they remain susceptible to **hallucinations**: responses that appear plausible while being incorrect, misleading, or entirely fabricated.

As AI systems become increasingly autonomous, **trustworthiness is emerging as a fundamental software quality attribute for AI systems.** This project investigates how software engineers can design AI systems that recognise the limits of their own knowledge before presenting information to users.

Rather than attempting to eliminate hallucinations entirely, this assignment explores whether agent-based validation architectures can detect, prevent, or appropriately communicate uncertainty before incorrect information reaches the user.

The project is therefore framed as an empirical software engineering investigation into AI trustworthiness.

1.1 Research Statement

> **A Confident Agent is not one that is always correct. It is one that knows when it should not be trusted.**

1.2 Research Question

> Can an agent-based architecture detect or prevent unacceptable hallucinations before they reach the user?

1.3 Scientific Hypothesis

> Agent-based validation architectures equipped with explicit confidence estimation can significantly reduce unacceptable hallucinations compared with an equivalent baseline Large Language Model operating without validation.

1.4 Guiding Principles

Every project undertaken in this course should satisfy the following principles.

1. **No intervention without a baseline.**

   Every proposed solution must be compared against an experimentally justified baseline.

2. **No claim without evidence.**

   Every conclusion must be supported by appropriate experimental evidence.

3. **No conclusion without discussing validity.**

   Every experimental result has limitations. Students are expected to critically discuss threats to the validity of their conclusions.

1.5 Definitions

Confident Agent

An AI system that attempts to detect, prevent, or communicate unacceptable uncertainty before information is presented to the user.

Hallucination Benchmark

A curated collection of prompts designed to induce unacceptable hallucinations within a chosen application domain. The benchmark serves as the primary instrument for evaluating hallucination mitigation strategies.

Baseline Model

The unmodified Large Language Model against which the proposed intervention is evaluated.

Confidence Metric

A quantitative estimate describing the likelihood that a generated response is sufficiently trustworthy for presentation to a user.

Validation Pipeline

The collection of agents, tools, retrieval mechanisms, and verification strategies used to improve AI trustworthiness.

2 Introduction

Software engineering has traditionally focused on constructing systems that are correct, reliable, maintainable, secure, and efficient. The rapid adoption of Large Language Models (LLMs) has introduced a new engineering challenge. Unlike conventional software systems, LLMs may produce responses that are plausible, convincing, and entirely incorrect. These hallucinations are often difficult for users to detect because they resemble correct answers and are frequently presented with unwarranted confidence.

As generative AI becomes increasingly integrated into software engineering, scientific research, education, healthcare, finance, and public decision making, the ability to recognise uncertainty is becoming as important as the ability to generate information. Consequently, **trustworthiness is emerging as a fundamental software quality attribute for AI systems.**

This assignment investigates how software engineers can design AI systems that recognise the limits of their own knowledge. Rather than attempting to eliminate hallucinations entirely, students are challenged to investigate whether carefully engineered validation architectures can detect, prevent, or appropriately communicate uncertainty before incorrect information reaches a user.

Unlike traditional software engineering projects, the objective of this assignment is not simply to develop an AI-powered application. Instead, students will conduct an empirical software engineering investigation into AI trustworthiness. Each project will construct an experimental benchmark, establish a measurable baseline, implement an intervention, evaluate its effectiveness, and critically analyse the validity of the resulting conclusions.

Throughout the project, students are encouraged to adopt the mindset of researchers rather than application developers. The software they construct serves as the experimental apparatus; the benchmark serves as the measuring instrument; the statistical analysis provides the evidence; and the final report constitutes the scientific contribution.

By the completion of this project, each group should have contributed not only a working *Confident Agent*, but also reproducible experimental evidence that advances our understanding of hallucination detection and prevention within a chosen application domain.

Throughout this project, emphasis is placed on thoughtful design rather than implementation alone. Students are expected to justify the design of their benchmark, experimental

methodology, and agent architecture before considering implementation details.

3 Project Goals

The primary goal of this project is to investigate whether agent-based validation architectures can improve the trustworthiness of Large Language Models through the detection and prevention of unacceptable hallucinations.

Unlike traditional software engineering projects, the primary deliverable of this assignment is not merely a functional software application. Instead, the project is intended to emulate an empirical software engineering investigation in which students design, implement, evaluate, and critically analyse an intervention intended to improve the trustworthiness of AI systems.

Each project must therefore contribute both an engineering artefact and experimental evidence supporting (or refuting) the effectiveness of the proposed approach.

The objectives of this project are divided into three complementary categories.

3.1 Educational Objectives

Upon successful completion of this project, students should be able to

- design and implement multi-agent AI systems;
- integrate modern AI engineering techniques including tool calling, retrieval, external APIs, and validation pipelines;
- construct domain-specific evaluation benchmarks;
- design controlled experiments for evaluating AI systems;
- apply appropriate statistical methods when analysing experimental results;
- critically evaluate the trustworthiness of AI-generated information.

3.2 Engineering Objectives

Students are expected to design and implement a **Confident Agent** capable of improving the trustworthiness of an existing Large Language Model through one or more hallucination mitigation strategies.

The architecture should include appropriate mechanisms for confidence estimation, validation, verification, retrieval, agent collaboration, or other experimentally justified techniques.

Students are encouraged to investigate novel architectures where appropriate.

3.3 Scientific Objectives

Each project should investigate a clearly defined research question relating to hallucination detection or prevention within a chosen application domain.

Projects are expected to

- establish an experimentally justified baseline;
- construct or obtain an appropriate benchmark;
- evaluate a proposed intervention;
- analyse the resulting evidence;
- discuss the validity of the conclusions drawn.

3.4 Primary Deliverable

The primary deliverable of this assignment is not an AI application.

The primary deliverable is experimentally defensible evidence that the proposed intervention improves (or does not improve) the trustworthiness of an AI system within a chosen application domain.

4 Experimental Methodology

Every project undertaken in this course shall follow the same experimental methodology. The purpose of this methodology is to ensure that all proposed interventions are evaluated fairly, reproducibly, and using comparable experimental evidence.

Students are expected to investigate whether a proposed *Confident Agent* can improve the trustworthiness of a Large Language Model within a chosen application domain.

4.1 Experimental Overview

Every project follows the same empirical software engineering workflow. Students first measure the behaviour of an unmodified Large Language Model using a carefully constructed benchmark. They then design and implement an intervention intended to improve trustworthiness before repeating the identical evaluation. Finally, they analyse the results and critically evaluate the validity of their conclusions.

This workflow ensures that improvements are supported by experimental evidence rather than anecdotal observations.

4.2 Phase 1 – Choose an Application Domain

Each group shall identify a domain in which hallucinations may have undesirable consequences.

Typical examples include, but are not limited to,

- academic citations;
- software APIs;
- software engineering;
- medical information;
- legal information;
- education;
- financial advice;
- historical knowledge.

Groups may propose alternative domains subject to lecturer approval.

4.3 Phase 2 – Construct a Hallucination Benchmark

Each project shall construct (or justify the use of) a benchmark designed to expose hallucinations within the chosen application domain.

The benchmark should contain sufficient prompts to evaluate the behaviour of both the baseline system and the proposed intervention.

The benchmark itself constitutes one of the primary research artefacts produced by the project.

A well-designed Hallucination Benchmark should be representative of the chosen application domain, contain prompts that consistently expose the behaviour under investigation, permit objective evaluation wherever possible, and be sufficiently documented to allow independent reproduction of the reported experiments.

4.4 Phase 3 – Establish a Baseline

The benchmark shall first be executed using an unmodified Large Language Model.

This establishes the experimental control against which all subsequent improvements will be measured.

Students should record quantitative measurements appropriate to their chosen domain.

4.5 Phase 4 – Design a Confident Agent

Groups shall design and implement an agent-based architecture intended to improve the trustworthiness of the baseline system.

The intervention may employ one or more techniques including

- retrieval;
- verification;
- agent collaboration;
- tool calling;
- confidence estimation;
- external knowledge sources;
- other experimentally justified techniques.

The choice of architecture must be justified experimentally.

Confidence estimation is inherently application dependent. Students should therefore select and justify an approach appropriate to their chosen domain. Confidence may be estimated using one or more techniques, including self-verification, consensus among multiple agents, external validation, retrieval-based verification, tool-assisted validation, or other experimentally justified approaches. No single confidence estimation strategy is prescribed.

4.6 Phase 5 – Comparative Evaluation

The identical benchmark shall be executed using the completed Confident Agent.

Students should avoid modifying benchmark prompts after baseline evaluation has been completed.

Students should consider both successful hallucination mitigation and unintended side effects. For example, an intervention that correctly prevents hallucinations may also incorrectly reject valid responses or communicate unnecessary uncertainty. Where appropriate, the evaluation should therefore consider both incorrect acceptance and incorrect rejection of responses.

4.7 Phase 6 – Statistical Analysis

Experimental results shall be analysed using appropriate statistical techniques.

The objective is to determine whether the proposed intervention produced a meaningful improvement over the established baseline.

4.8 Phase 7 – Threats to Validity

Every report shall critically discuss limitations of the experimental methodology.

Typical considerations include benchmark bias, evaluator bias, overfitting, model drift, confidence calibration, and the generalisability of the reported results.

Students should also consider practical engineering trade-offs introduced by their proposed solution. Improvements in trustworthiness may be accompanied by increased computational cost, latency, API usage, energy consumption, or implementation complexity. Such trade-offs should be acknowledged where they may influence the practical applicability of the proposed approach.

5 Agent Architecture

This assignment investigates the use of agent-based architectures to improve the trustworthiness of Large Language Models. Students are free to design any architecture appropriate to their chosen application domain, provided that the resulting system can be evaluated experimentally.

Although no specific implementation is prescribed, most successful solutions are expected to distinguish between the generation of candidate responses and their subsequent validation.

A typical architecture may therefore include a coordinating *Chief Agent* responsible for interacting with the user together with one or more specialised agents responsible for activities such as

- information retrieval;
- verification;
- confidence estimation;
- external tool usage;
- citation validation;
- API validation;
- fact checking;
- domain-specific reasoning.

Alternative architectures are encouraged provided they are experimentally justified and consistent with the research objectives of the project.

6 Project Timeline

This project will be completed over the course of the semester according to the milestones outlined below. Each milestone builds upon the previous one and contributes towards the final project submission.

Milestone	**Expected Outcome**
Project Proposal	Application domain selected and approved.
Benchmark Design	Hallucination benchmark completed and documented.
Baseline Evaluation	Baseline LLM evaluated using the benchmark.
System Implementation	Confident Agent implemented and demonstrated.
Experimental Evaluation	Baseline and intervention compared using the same benchmark.
Final Submission	Report, source code, benchmark, and presentation submitted.

Detailed submission dates and assessment deadlines will be communicated during the semester and published on the course learning management system.

Students are encouraged to begin benchmark construction as early as possible, since all subsequent stages of the project depend upon the availability of a well-designed benchmark.

7 Assessment

Assessment for this project will consider both the quality of the implemented system and the quality of the scientific investigation undertaken by the group.

Marks will be awarded for the design and implementation of the proposed Confident Agent, the quality of the experimental methodology, the analysis of the resulting evidence, and the clarity of the final report and presentation.

The assessment will be based on the following components.

Component	**Description**
Project Proposal	Clearly defines the application domain, research question, and proposed approach.
Benchmark	Quality, relevance, completeness and justification of the benchmark used during the investigation.
Confident Agent	Design, implementation and technical quality of the proposed agent architecture.
Experimental Evaluation	Correct execution of the experimental methodology, including baseline evaluation and comparative analysis.
Statistical Analysis	Appropriate analysis and interpretation of the experimental results.
Threats to Validity	Critical discussion of the limitations of the methodology, benchmark, implementation and conclusions.
Report	Organisation, technical quality, scientific writing and reproducibility of the investigation.
Presentation	Professional presentation and demonstration of the completed project.

Detailed assessment criteria and mark allocations will be provided in the accompanying assessment rubric.

Projects will primarily be assessed on the quality of the evidence presented rather than on whether a particular intervention proved successful. Well-executed investigations that produce carefully justified negative results may therefore receive full credit.

8 Reporting Requirements

Each group shall submit a written technical report documenting the design, implementation and evaluation of their project. The report should be written in a clear, professional style and should present the project as an empirical software engineering investigation.

Unless otherwise instructed, the report should include the following sections.

1. Introduction
2. Research Question
3. Application Domain
4. Benchmark Design

5. Baseline Evaluation
6. Confident Agent Design
7. Experimental Methodology
8. Experimental Results
9. Statistical Analysis
10. Threats to Validity
11. Conclusion

The report should present sufficient information to allow the reader to understand the investigation, reproduce the experimental methodology, and critically evaluate the conclusions.

All claims should be supported by appropriate evidence obtained during the investigation.

Figures, tables and diagrams should be used where appropriate to improve clarity.

Groups are encouraged to discuss limitations of their work and identify opportunities for future improvement where appropriate.

In addition to the written report, each group shall submit the source code, benchmark, supporting artefacts and any additional material required to reproduce the reported results.

A Appendix

A.1 Suggested Application Domains

The following examples are provided to illustrate the types of application domains suitable for investigation. Groups are encouraged to propose alternative domains subject to lecturer approval.

- Academic citation generation and validation
- Software API recommendation and verification
- Medical information retrieval and validation
- Legal information systems
- Financial information assistants
- Educational tutoring systems
- Historical fact verification
- Scientific literature assistants
- Requirements engineering assistants
- Domain-specific knowledge assistants

A.2 Suggested Hallucination Types

Projects may investigate one or more forms of hallucination, including

- fabricated references;
- incorrect factual claims;
- invalid software APIs;
- fabricated quotations;
- incorrect mathematical reasoning;
- unsupported recommendations;
- fabricated tools or libraries;
- inconsistent reasoning;
- domain-specific misinformation.

A.3 Suggested Technologies

Students may select technologies appropriate to their proposed architecture. Examples include

- Large Language Models
- Agent frameworks
- Retrieval-Augmented Generation (RAG)
- Model Context Protocol (MCP)
- External APIs
- Vector databases
- Knowledge graphs
- Traditional databases
- Search engines

The choice of technology will not be assessed directly. Assessment will focus on the quality of the proposed solution, its experimental evaluation, and the evidence presented in support of the reported conclusions.

A.4 Academic Integrity

All submitted work must comply with the University's policies on academic integrity.

Students are expected to acknowledge the use of generative AI tools, external datasets, software libraries and other third-party resources where appropriate.

The submitted work must accurately represent the contribution of the group members and must be reproducible from the supplied artefacts.

References